

AXEL v2 – Phase 2 Product Requirements Document

Data Ingestion, Point-in-Time Data, Backtester, Validation and Strategy Registry

Build-ready implementation specification

1 Purpose and Source Requirements

Phase 2 creates the research/validation substrate for AXEL. The approved build plan requires Alpaca/Polygon bars, FRED, SEC EDGAR and one news source, all point-in-time; a backtester with fees, slippage, market calendars and walk-forward splits; validation reports containing expectancy R , Sharpe, maximum drawdown, trade count, parameter sensitivity and multiple-testing adjustment; and two or three baseline strategies including trend-following and mean-reversion. Exit requires reproduction of a known textbook result and rejection of a deliberately overfit strategy.

2 Architectural Rules

- Data/research code never calls execution or holds broker credentials.
- The deterministic Trading core and Watchdog remain LLM-free; CI must enforce this.
- Research evidence crosses boundaries only through typed/versioned records.
- All internal timestamps are timezone-aware and normalized to UTC; source-local timestamps are retained.
- A historical query at time t may use only information publicly available at or before t .
- Every dataset, feature, strategy and validation run must be reproducible by version/hash.

3 Scope

3.1 In scope

provider adapters, raw ingestion, normalization, provenance, point-in-time joins, calendars/corporate actions, deterministic feature pipeline, event-driven backtester, execution assumptions, costs/slippage, walk-forward/holdout validation, robustness, multiple-testing accounting, baseline strategies, strategy registry, reports and tests.

3.2 Out of scope

live trading, broker execution, LLM analysts, paper-soak operations, other market sections and autonomous deployment.

4 Repository Responsibilities

Recommended modules:

axel/data/providers/	axel/data/ingest/
axel/data/normalize/	axel/data/point_in_time/
axel/data/calendars/	axel/data/schemas/
axel/backtest/engine/	axel/backtest/costs/
axel/backtest/slippage/	axel/backtest/metrics/
axel/backtest/splits/	axel/backtest/reports/
axel/strategies/	axel/ml/features/
axel/ml/validation/	tests/data/
tests/replay/	tests/integration/

Equivalent existing paths are acceptable if the boundaries remain intact.

5 Provider Interfaces

Each provider adapter exposes a common interface and converts provider-specific formats to canonical records.

5.1 Market

Alpaca/Polygon-compatible OHLCV: symbol, venue, timestamp, OHLC, volume, source timestamp, adjustment status, ingestion timestamp, schema/version and source hash.

5.2 FRED

series ID, observation date, value, release/publication timestamp when available, revision/vintage metadata and source provenance. Backtests use the vintage available at the decision time.

5.3 EDGAR

issuer ID, accession, form, filed timestamp, period end, source hash, extraction version and normalized fields. Filing information becomes usable only at actual public filing time.

5.4 News

article ID, publisher, publication time, retrieval time, title/body reference, entity mappings, source hash and provenance. External text is data, never instructions.

6 Canonical Data Contract

Every normalized record contains immutable source ID/name, event/publication time, ingestion time, entity/instrument ID, schema version, source payload hash, normalization version, quality flags and enough provenance to reconstruct the source. Raw payloads should be retained immutably; normalized data is queryable.

7 Point-in-Time Integrity

For every feature:

$$available_at = \max(source_available_at, dependency_availability_times).$$

Joins must use availability time, not economic period date. A quarterly filing cannot be used on quarter-end if filed later. Revised macro data cannot replace the vintage historically available. Mandatory tests: future filing injection, future-news injection, macro-vintage/revision, split/corporate-action, timezone/session boundary, duplicate and out-of-order event tests.

8 Ingestion Pipeline

Fetch → validate transport → persist immutable raw payload/hash → normalize → validate → deduplicate → attach provenance/quality → versioned storage → metrics/error reporting. The pipeline must be restartable and idempotent: repeating an interval must not create duplicate logical records.

9 Data Quality

Validate missing/invalid timestamps, impossible OHLC relationships, negative volume, duplicates, gaps, symbol mapping, timestamp ordering, stale feeds and provider disagreement. Invalid records must fail closed for validation rather than being silently interpolated.

10 Calendars and Corporate Actions

Implement exchange sessions, holidays, early closes, market-open/close state and explicit split/dividend adjustment policy. Research-adjusted prices and execution-realistic prices must be distinguishable; double adjustment is forbidden.

11 Feature Dataset

Feature rows contain entity, event time, availability time, feature-set version, source versions, values, missingness indicators and quality status. Initial families: returns/momentum, rolling volatility/range, volume/liquidity, cross-sectional ranks, trend state, mean-reversion distance, point-in-time fundamentals, point-in-time macro and news-derived metadata/sentiment. Every feature has documented intent and tests.

12 Backtester

The engine must be event-driven and deterministic. Inputs: versioned dataset, strategy version, capital, calendar, fees, slippage, position constraints and split configuration. Outputs: orders, fills, positions, cash, equity, exposure, costs and P&L, plus complete metrics. Execution must define timing, fees, spread/slippage, partial fills where relevant, market hours, cash constraints, deterministic IDs and rejected orders. It must never assume a same-bar fill unless the data and strategy explicitly support it.

13 Walk-Forward Validation

Use rolling development/training, validation and forward OOS windows, plus an untouched final holdout never used for tuning. Record all boundaries and embargo/gap rules. Any fitted preprocessing is fitted separately inside each fold.

14 Validation Metrics and Gates

Reports must contain trade count, gross/net P&L, expectancy in R , Sharpe with assumptions, maximum drawdown, win rate, payoff distribution, turnover, fees/slippage, exposure/concentration, regime/time slices, parameter sensitivity, split definitions, variant count and multiple-testing/deflated-Sharpe information. Committed defaults: at least 100 OOS trades; net expectancy approximately $0.1R$ – $0.5R$; default net OOS Sharpe above 0.7; maximum drawdown within section tolerance; $\pm 20\%$ parameter perturbations must not flip expectancy sign; evidence across at least two regimes. Sustained results above about $1R$ expectancy trigger a leakage audit, not automatic promotion.

15 Baseline Strategies

Implement at least:

1. trend-following/momentum;
2. mean-reversion;
3. optional third deterministic baseline where useful.

Each strategy defines signal, entry, exit, parameters, data dependencies, invalidation and version metadata. Baselines are controls, not guarantees.

16 Strategy Registry

Fields: strategy ID, section, version, code/data dependencies, parameters, risk assumptions, validation run IDs, author, timestamps and status. Lifecycle:

DRAFT $\rightarrow$ *BACKTESTED* $\rightarrow$ *VALIDATED* $\rightarrow$ *PAPER_ELIGIBLE* $\rightarrow$ *ACTIVE*

with explicit REJECTED/RETIRED states. Phase 2 cannot grant live authority.

17 Validation Report Artifact

Every report must include run ID, code commit, dataset/feature/strategy hashes, split definitions, costs/slippage, all metrics, OOS trade count, robustness, multiple-testing accounting, limitations, pass/fail decision and exact reproducibility reference.

18 Testing Requirements

18.1 Unit

provider parsing, normalization, timestamp conversion, calendars, features, costs, slippage, metrics, split generation and registry transitions.

18.2 Property

no future data enters features; ingestion is idempotent; record ordering does not change results; splits never overlap; execution assumptions cannot create impossible fills.

18.3 Integration

provider fixtures, database round-trip, fixed-data replay, full backtest and report generation.

18.4 Golden

a textbook strategy must reproduce a known result; a deliberately overfit strategy must fail the admission gate.

19 Observability and Security

Expose ingestion lag/errors, missing data, provider health, backtest run IDs/duration, dataset versions, validation status and failed checks. Logs must contain correlation IDs and no secrets. Research jobs cannot modify risk limits, execution, watchdog state or broker credentials.

20 Assignable Work Packages

ID	Module	Deliverable
P2-DATA	Providers/ingestion	Adapters, raw persistence, normalization, idempotency
P2-PIT	Point-in-time	Availability-aware joins and leakage tests
P2-FEAT	Features	Versioned deterministic feature pipeline
P2-BT	Backtester	Event engine, fills, costs, slippage
P2-VAL	Validation	Walk-forward, holdout, robustness, multiple testing
P2-STRAT	Strategies	Baselines and registry lifecycle
P2-REP	Reports	Reproducible validation artifacts
P2-QA	Quality	Golden, property, integration and replay suite

21 Developer Boundaries

Developers must not alter deterministic risk math, execution approval semantics, watchdog behavior, broker credentials or LLM trust boundaries as part of Phase 2. Cross-boundary changes require separate review.

22 Definition of Done

1. All required providers have deterministic fixtures.
2. Point-in-time tests pass.
3. Backtester reproduces the reference result.
4. Fees, slippage and calendars are active.
5. Walk-forward and untouched holdout are enforced.
6. At least two baseline strategies are registered.
7. The overfit strategy is rejected.
8. Reports are reproducible and versioned.
9. No unresolved leakage/provenance/duplicate defects remain.
10. CI passes the full Phase 2 suite.
11. A strategy is traceable from raw source through feature, backtest, validation and registry state.

23 Acceptance Test

From a clean environment, load fixed market/EDGAR/FRED/news fixtures; create point-in-time features; run a baseline with costs/slippage and calendar rules; generate a walk-forward report; reproduce the known result; reject the overfit strategy; and verify that every artifact identifies exact source, feature, strategy and code versions.

24 Handoff

Phase 2 feeds Phase 3 with trusted point-in-time data, feature versions and validated Stocks playbook entries. Phase 3 must consume these typed/versioned outputs rather than duplicating data logic.